AI Learning Companion

Team 7 Needfinding

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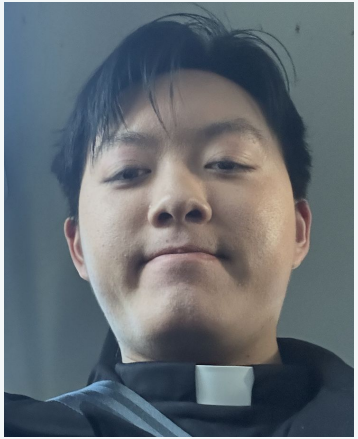

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Project Name: Reason Lab

“Better Learning Starts With You “

Why "*Reason Lab*"

- Reason
  - Our product prioritizes your own reasoning. Requiring you to think and attempt the problem before aiding your thinking
- Lab
  - Low stakes + Experimental environment
  - Meant to be a place where you can try over and over until you finally understand
- Reason + Lab
  - Learning through iteration + AI built off the user

01

Problem and Solution Overview

Problem Overview

- Audience / Users
  - College + High School Students dealing w/ complex and difficult to understand coursework

- Problem
  - Nowadays with AI so readily available, many are using it as a shortcut.
  - This use of AI is taking away from in-depth learning, and can

What is our solution?

- Reason Lab enforces active learning by forcing the users to attempt a solution before getting any aid
  - Makes its users strengthen their own reasoning rather than an instant solution
  - Will give adaptive guidance and guide the user, forcing a more durable understanding of material
  - Through adaptive, visual guidance, ReasonLab responds to user attempts and misconceptions, helping learners refine their thinking and build a more durable conceptual understanding over time.

02

Market Research

Other Competitors

Khan Mingo	Overview: Khan Academy's AI tutor that helps students finish their assignments.
	Compare: Similar guiding / step by step system but isn't strictly required. Can still ask for the solution
Socratic	Overview: An AI-driven homework helper that lets students take a photo of a question or type it in and receive explanations and step-by-step breakdowns.
	Compare: Surface level similarity, in that its education focus. Gives instant answers with an explanation which is the opposite of our own solution.
ChatGPT	Overview: A widely used LLM thats extremely flexible and quick, oftenly used to answer and explain problems in education
	Compare: Completely self guided, leaving the learning discipline all to the users.

Feature / Capability	ReasonLab	KhanMingo	ChatGPT-5.2	Socratic
User-friendly	Yes	Yes	Yes	Yes
Guides, not answers	Yes	Yes	User-Initiated	User-Initiated
Visual explanations (diagrams / flowcharts)	Yes	No	Limited	Limited
Encourages independent thinking	Yes	Yes	No	No
Answer-with-design (structured responses)	Yes	Yes	No	No
Supports complex reasoning processes	Yes	Limited	Yes	Limited

03

Tasks

Task 1- Simple (0:58-1:01)

Review / Peer Review

What the user wants to do:
A user would like to verify a simple assignment (Multiple Choice or SAQ) and is sending it through “Reason Lab” to make sure they understood and correctly answered the assignment

Why this is Simple:

- Introductory task
- A topic the user already has an understanding of
- Minimal interaction, sending in already finished work
- Correct or Incorrect, quick response (i.e. multiple choice)

Task 2- Moderate (1:02-1:29)

Iterating/Revising an Answer

What the user wants to do:
A user wants to work on a more subjective problem. Will have to send it a paragraph of more in order to answer a question, and have to interact w/ “Reason Lab” to improve the paragraph.

Why this is Moderate:

- The user has a partial understanding of the topic
- Moderate cognition, interpreting and applying feedback.
- Moderate interaction, repeatedly sending progress back, and getting feedback

Task 3 - Moderate

Assess their understanding

What the user wants to do:
A student will explain a topic to “Reason Lab” and evaluate their understanding of the topic

Why this is Moderate:

- Requires reflection and critical thinking, rather than a simple confirmation
- No singular solution / correct answer, more so reframing knowledge the user already knows
- Heavy engagement with the AI, requiring repeated explanations, simplifications, and iterations

Task 3 - Complex (1:30-1:45)

Learning an entirely new concept

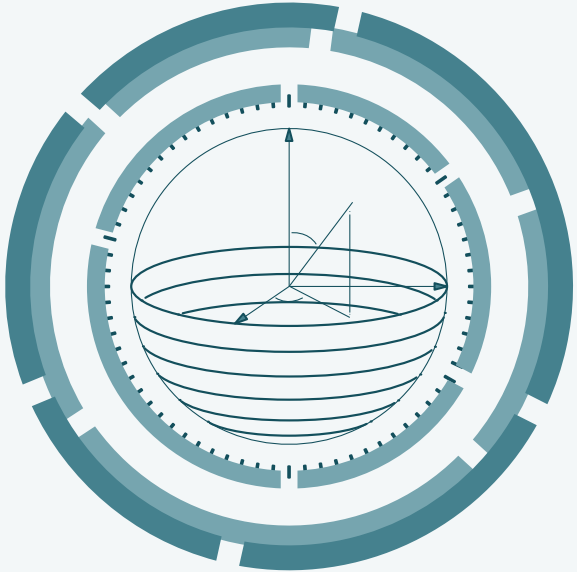
What the user wants to do:
A student wants to learn a completely new set of coursework or a concept they have never encountered before.

Why this is Complex:

- High cognitive effort and sustained engagement
- Requires constructing understanding rather than verifying or refining existing knowledge
- Involves uncertainty, decision-making, and iterative reasoning
- Intended to develop critical thinking, iteration, and deeper mastery

04

Storyboards for the videos



Storyboard scenes

Frame 1 – Opening / Hook

Frame 2 - Solution Intro

Frame 3 – Task 1: Simple Review / Peer Review

Frame 4 – Task 2: Iterating / Revising an Answer

Frame 5 – Task 3: Learning a New Concept

Frame 6 – Outcome / Impact

Frame 1 – Opening / Hook (0:00–0:33)

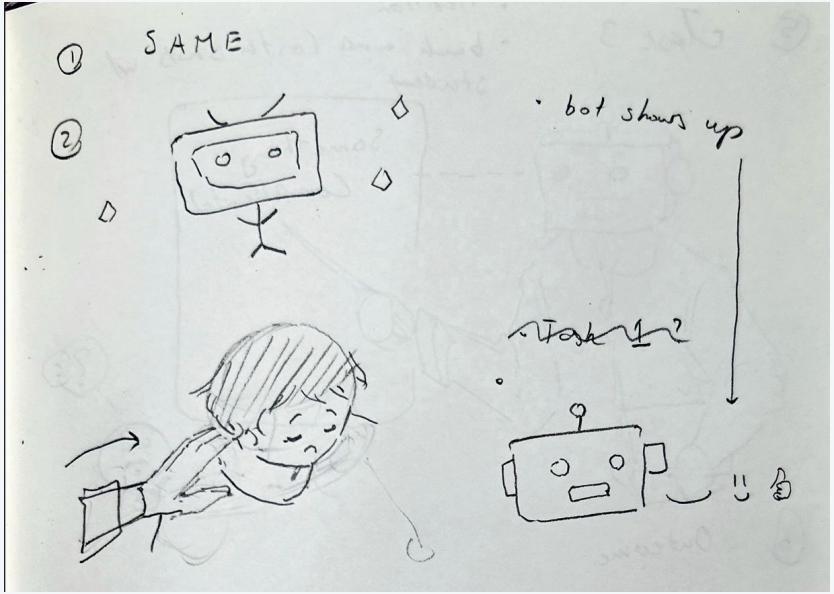
- **Visual:** Student stressed over a complex assignment, staring at a screen. Gives up and decides to just ChatGPT it (“Man I’m not doing allat”)
  a. 5 weeks later, on the test
  b. Fails + Distraught

- **Purpose:** Establish problem context.

Frame 2 – Solution Intro (0:34–0:57)

- **Visual:** Appearance of the Reason lab AI bot
  a. Hand on shoulder
  b. Pan to the robot doing a thumbs up

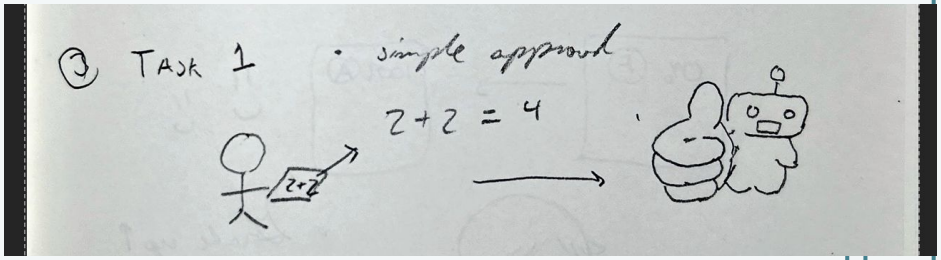
- **Purpose:** Begin introducing our solution to the problem



Frame 3 – Task 1: Simple Review / Peer Review (0:58–1:01)

- **Visual:** Student solving a simple math problem
  a. Student hands paper
  b. Robot gives approval

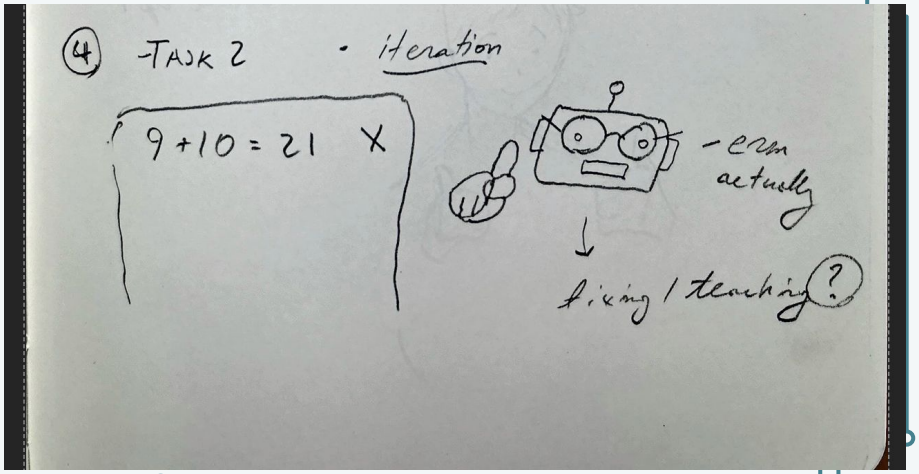
- **Purpose:** Show the base case where the user gives an input/complete work and has their understanding reviewed



Frame 4 – Task 2: Iterating / Revising an Answer (1:02–1:29)

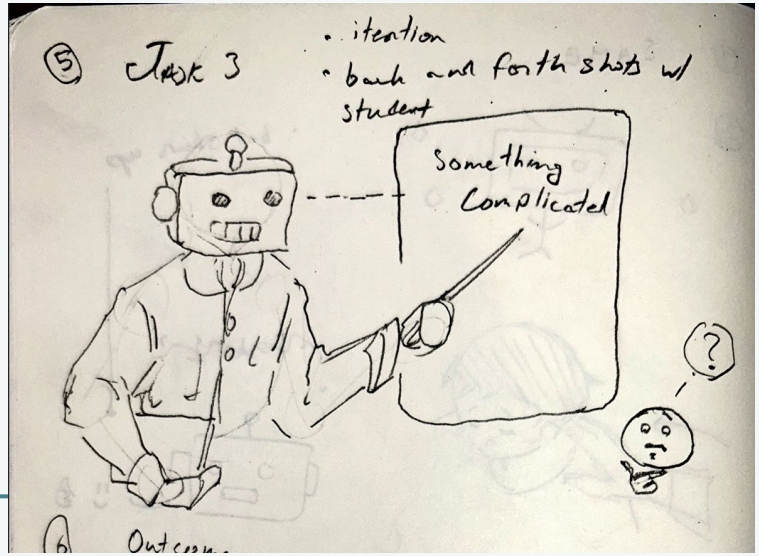
- **Visual:** Incomplete/Incorrect work that requires feedback from the bot
  a. Incorrect work on the paper
  b. Robot advising the student
  c. Display understanding

- **Purpose:** Display how the bot will fix/adjust your work while letting the user do most of the thinking



Frame 5 – Task 3: Learning a New Concept (1:30-1:45)

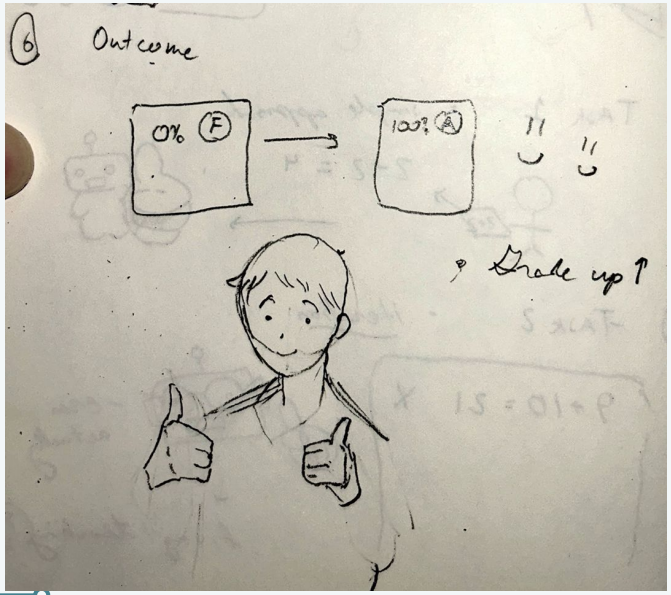
- **Visual:** Bot teaching a complex issue
  a. After receiving input/questions from the user, bot will begin teaching
  b. Repeated back and forth shots between bot and user
- **Purpose:** Displays how the bot is useful in learning via constant user participation



Frame 6 – Outcome / Impact (1:46–1:56)

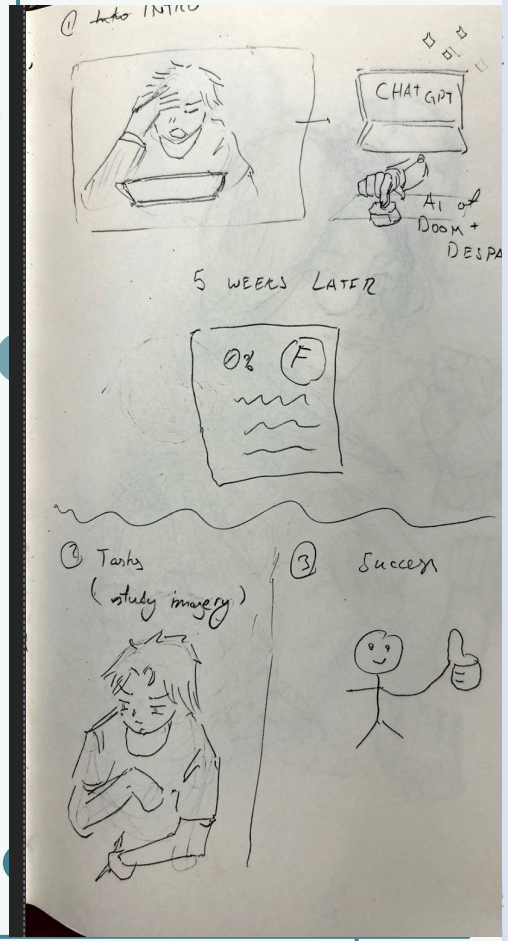
- **Visual:** Student improving their grade
  a. 0 → 100%
  b. Show student expression.

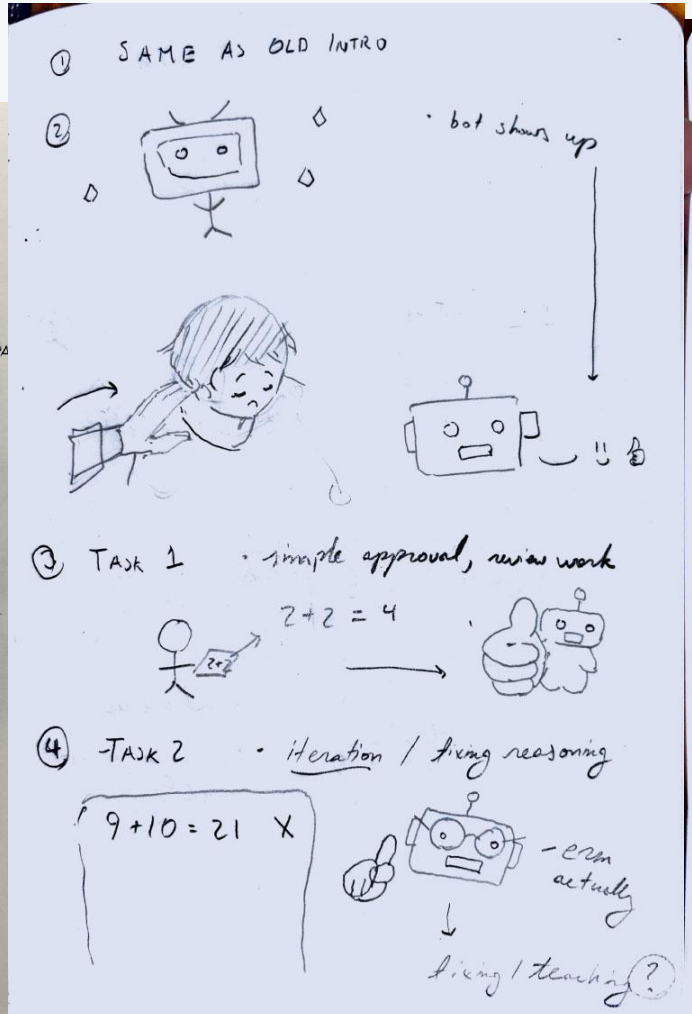
- **Purpose:** Display the difference of using ReasonLab and critical thinking in learning

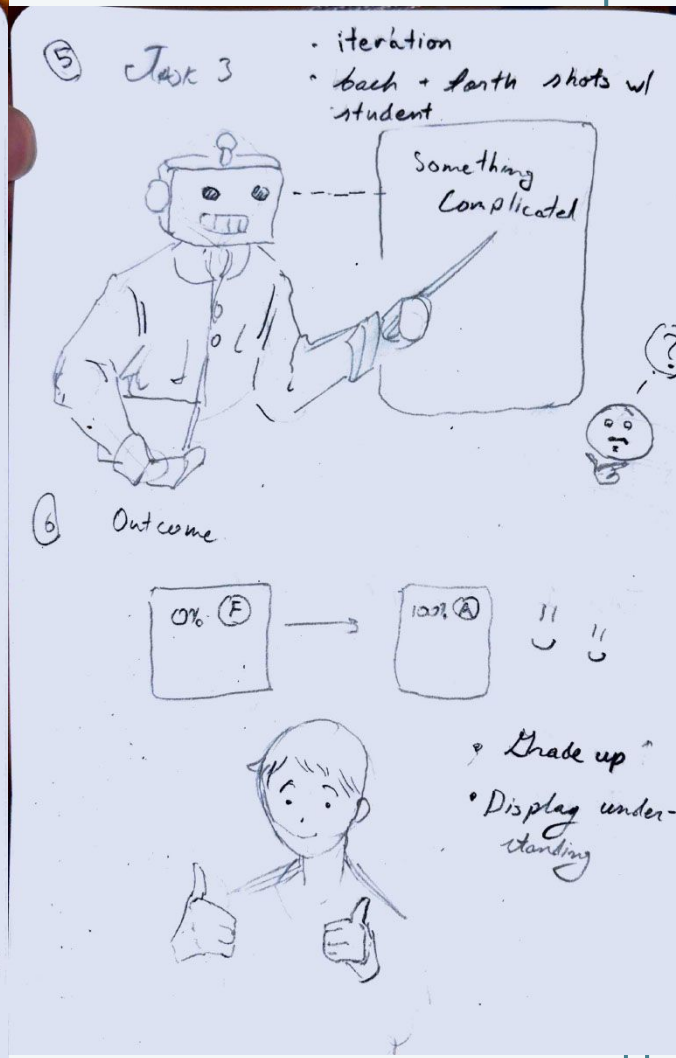


Video

05 Appendix


① Intro INTRO
CHAT GPT
AI of Doom + DESPA
5 WEEKS LATER
0% F
② Tasks (study imagery)
③ Success


① SAME AS OLD INTRO
② · bot shows up
③ TASK 1 · simple approval, review work
2+2 = 4
④ TASK 2 · iteration / fixing reasoning
9+10 = 21 X
- erm actually
fixing / teaching ?


⑤ Task 3
· iteration
· back + forth shots w/ student
Something Complicated
⑥ Outcome
0% F
100% A
· Grade up
· Display understanding